



আন্তর্জাতিক ইসলামী বিশ্ববিদ্যালয় চট্টগ্রাম
الجامعة الإسلامية العالمية شيتاغونغ
International Islamic University Chittagong

PROJECT REPORT

HOSPITAL MANAGEMENT SYSTEM

Course Code: CSE-3532

Course Title: Tools and Technologies for Internet Programming

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Overview

SmartCare is an advanced hospital management system built using Django and Django REST Framework (DRF), designed to improve patient care and hospital efficiency. It offers seamless doctor-patient interaction, secure user authentication with email confirmation, appointment scheduling with automated email alerts, and real-time search and filtering for doctors based on name or specialization.

How it is useful

SmartCare is a useful hospital management system that simplifies hospital tasks like appointment booking, doctor schedules, and patient data handling through automation. It allows patients to easily search for doctors, book appointments, and receive real-time email notifications. Doctors can manage appointments through a dedicated panel, while the responsive design ensures smooth use on both desktop and mobile devices. With secure authentication, including email verification and token-based access, user data is kept safe.

Why This Project

SmartCare is important because it improves the overall patient experience by making healthcare more accessible and organized. It saves time for both hospital staff and doctors by reducing manual workload. All data is securely stored and managed centrally, ensuring easy access and privacy. By modernizing healthcare operations with scalable technology, SmartCare helps bring hospitals into the digital age efficiently and safely.

Technologies Used:

Backend

- **Django** (*Python Web Framework*) — powering the backend APIs, models, and building logic
- **Django REST Framework (DRF)** — crafting clean and scalable RESTful APIs for the frontend
- **SQLite** — lightweight and efficient data storage solution

Frontend

- **HTML, CSS, JavaScript** — building interactive, responsive, and user-friendly web pages

Authentication & Security

- **Django's built-in User model & authentication system** — robust and secure user management
- **Token Authentication (DRF)** — ensuring secure API access and user sessions
- **Email Verification System** — confirming user accounts through email to keep it safe and sound

Tools & Environment

- **Git & GitHub** — seamless version control and team collaboration
- **Visual Studio Code** — IDE for coding and debugging

Design Process

Our project, **SmartCare - Hospital Management System**, was carefully designed and developed through the following steps to ensure a smooth, user-friendly, and efficient healthcare experience:

1. Requirement Gathering

We began by identifying the key needs of both patients and doctors — such as appointment booking, doctor search, profile management, and communication — to create a comprehensive feature list.

2. System Design & Architecture

We designed a modular system architecture separating the frontend and backend, using Django REST Framework for APIs and a responsive frontend built with HTML, CSS, and JavaScript for a clean UI.

3. Frontend Development

Created responsive web pages allowing users to search doctors, book appointments, manage profiles, and read blogs, with real-time API integration using JavaScript.

4. Backend App & Model Setup

The backend was initialized by creating Django apps for different modules (e.g., patients, doctors, appointments).

- Defined models for users, doctors, appointments, reviews, and blogs
- Created **serializers** to handle data conversion between models and APIs

5. Backend Development

Implemented RESTful APIs to handle authentication, appointment management, doctor profiles, and blog posting. Secure email verification and notification systems were integrated for smooth communication.

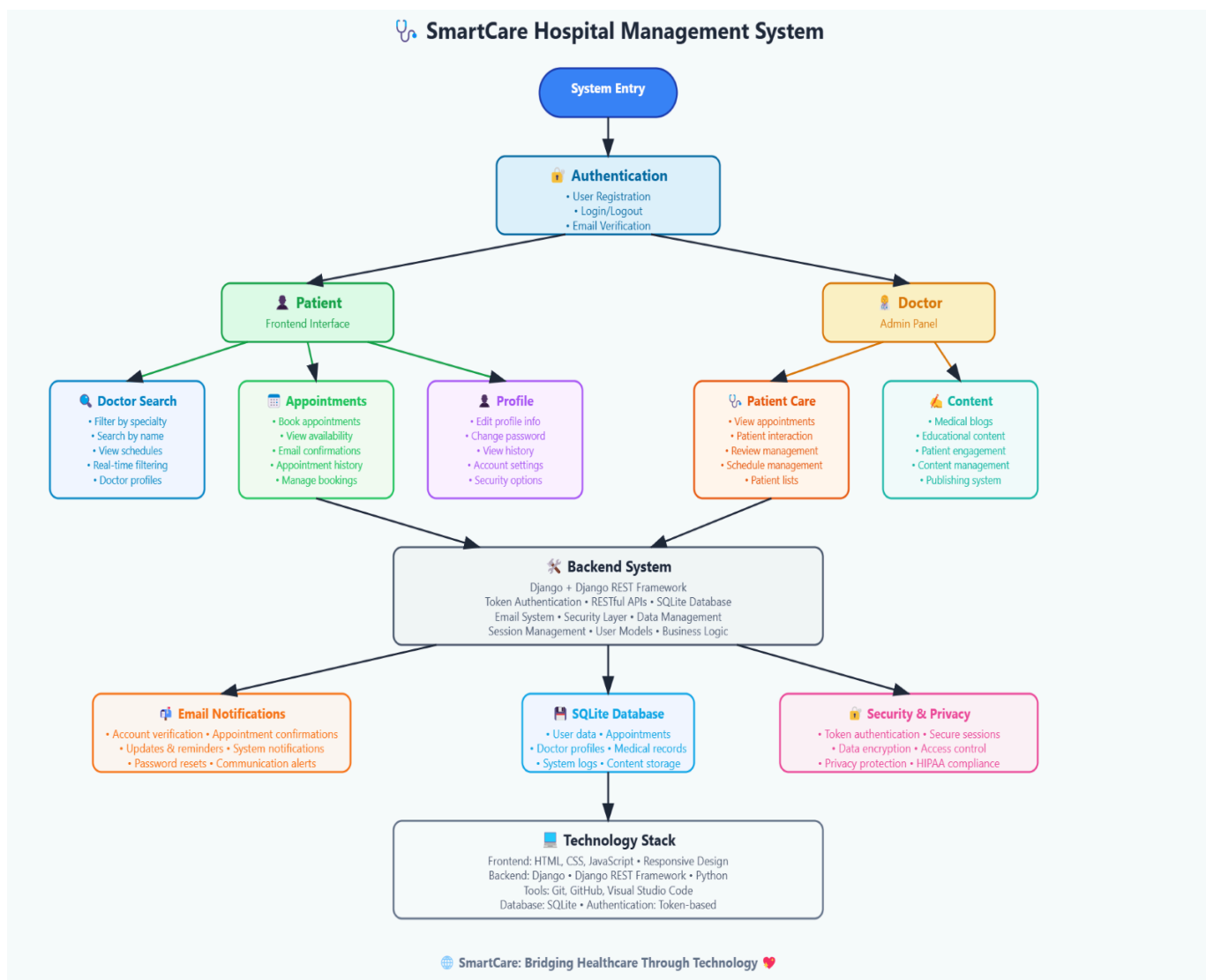
6. Testing & Debugging

Conducted thorough testing on both frontend and backend to ensure functionality, usability, and security. Bugs were fixed promptly to deliver a stable product.

7. Deployment & Documentation

Uploaded the complete project to GitHub with clear documentation, including README and project report, to assist users and future developers.

Flowchart



Future Plans & Next Phases

While **SmartCare** already offers essential features for hospital management, we have exciting ideas to expand and improve the system in future phases:

Phase 1: Enhanced User Experience

- **Live Chat Support** — Allow real-time chat between patients and doctors for faster consultations
- **Multi-language Support** — Make the platform accessible to a broader audience
- **Role-Based Dashboards** — Separate views for admin, patients, and doctors with personalized data
- **PDF Generation** — Allow users to download appointment confirmations and summaries in PDF format

Phase 2: Improved Security & Authentication

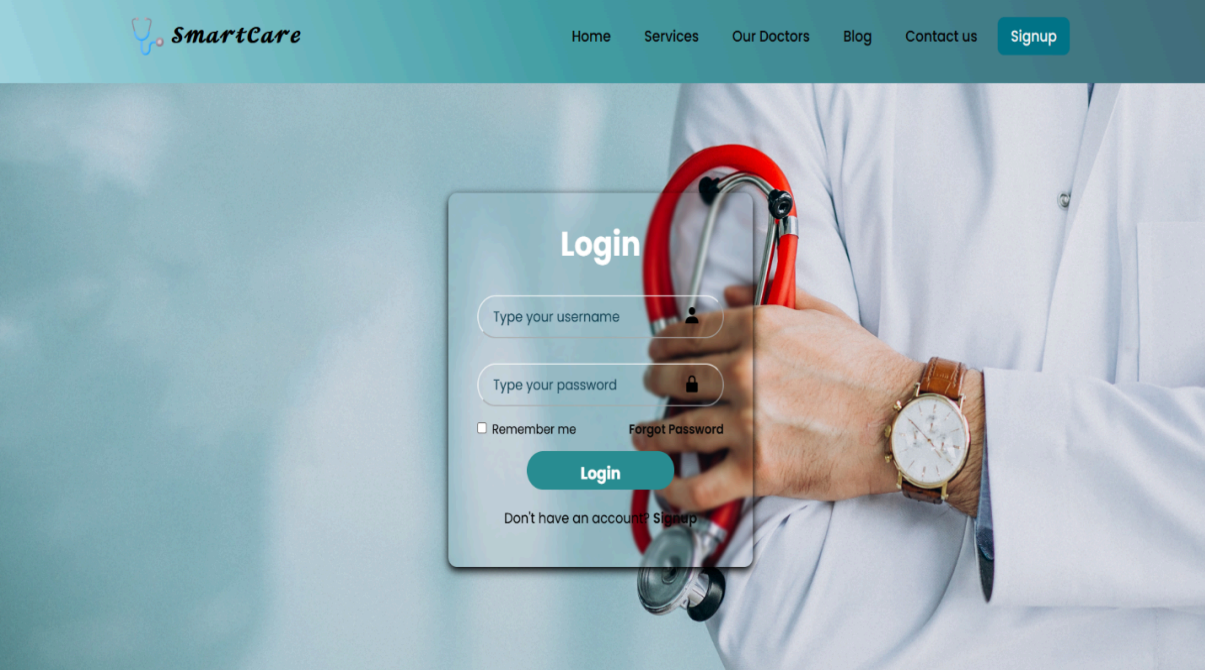
- **Two-Factor Authentication (2FA)** — For an added layer of account protection
- **Password Reset via Email/SMS** — To help users securely regain account access

Phase 3: Advanced Features & Automation

- **Analytics Dashboard** — Display patient visit stats, doctor performance, and service usage
- **AI-based Appointment Suggestions** — Recommend the best available doctors based on symptoms
- **E-Prescription Module** — Doctors can issue digital prescriptions viewable by patients

Phase 4: Mobile App Development

- **Android & iOS App** — A fully functional mobile version of SmartCare for on-the-go access
- **Push Notifications** — For reminders, confirmations, and real-time updates



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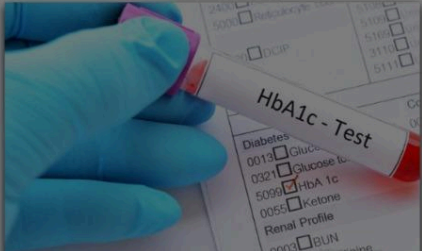


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Smartcare

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Conclusion

The **Hospital Management System** is an efficient solution that simplifies hospital appointment scheduling, improves communication between doctors and patients, and reduces manual administrative work. By combining features like secure login, email notifications and online consultations, the system enhances the overall healthcare experience. It is a scalable project with real-world applications and opportunities for future development.